Manan Tayal

Curriculum Vitae

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Evolution is a Tinkerer, So am I

Research Summary

My research lies at the intersection of **Control, Optimization, and Machine Learning**, with a focus on **safety-enforced decision-making** for autonomous systems. My Ph.D. thesis develops methods that integrate learning-based control with formal safety guarantees, ensuring constraint satisfaction and robust performance in uncertain, data-driven environments and validates them on real-world robotic platforms.

Keywords: Safe Reinforcement Learning, Control Barrier Functions (CBF), Hamilton-Jacobi (HJ) Reachability, Vision-Language-Action Models, Autonomous Robots.

Education

2021-2025 **Ph.D.**, *Indian Institute of Science (IISc)*, Bangalore
Thesis: Towards Trustworthy Autonomy in Uncertain Environments
Advisors: Prof. Shishir N.Y. Kolathaya & Pushpak Jagtap,
Safe and Performant Control of Autonomous Systems, CGPA: 9.30/10

2017-21 **B.Tech.**, *Indian Institute of Technology (IIT)* , Bombay
Engg.*CGPA: 9.13/10*

Research Experience

Jul'26 - Present **Research Scientist**, *Emergence AI India Labs*
Advisor: Dr. Prasenjit Dey

Jan'26 - Jun'26 **Visiting Researcher**, *Microsoft Research (MSR) India*
Advisor: Dr. Akshay Nambi

Nov'24 - Dec'25 **Student Researcher**, *Stanford University, USA*
Advisor: Prof. Somil Bansal

Apr'20 - May'20 **Research Intern**, *Siemens Technology and Services*, Bangalore, India
Advisor: Dr. Ram Padhy

Awards and Recognition

2021-2026 Recipient of the prestigious **Prime Minister's Research Fellowship (PMRF)**.

2024 Finalist of **Qualcomm Innovation Fellowship (QIF)**, India.

2018-2019 Awarded **AP grades** in two courses at IIT Bombay for exceptional performance.

2017 Secured **All India Rank 626** in JEE Advanced out of 1,59,540 candidates.

2017 Secured **All India Rank 289** in JEE Mains out of 10.2 lakh candidates.

2017 Recipient of **Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship.**

Selected Research Publications

(* - indicates Equal Contribution)

2026 **[10] Bridging Safety and Performance in Autonomous Systems using Offline Reinforcement Learning**, *[Paper]*
Manan Tayal*, Mumuksh Tayal*, Ravi Prakash
Accepted at Advances in Neural Information Processing Systems (NeurIPS).

[9] Safe Flow Q-Learning: Offline Safe Reinforcement Learning with Reachability-Based Flow Policies, *[Paper]*
Mumuksh Tayal*, **Manan Tayal***, Ravi Prakash
Reinforcement Learning Conference (RLC).

[8] Stochastic Neural Control Barrier Functions, *[Paper]*
Hongchao Zhang, **Manan Tayal**, Jackson Cox, Pushpak Jagtap, Shishir Kolathaya, Andrew Clark
Transactions on Automatic Control (TAC).

[7] V-OCBF: Learning Safety Filters from Offline Data via Value-Guided Offline Control Barrier Functions, *[Paper]*
Mumuksh Tayal*, **Manan Tayal***, Aditya Singh, Shishir Kolathaya, Ravi Prakash
Transactions on Machine Learning Research (TMLR).
AAAI Workshop on Deployable AI (DAI), 2026 (Oral).

[6] A Theoretical and Empirical Study on the Convergence of Adam with an Exact Constant Step Size in Non-Convex Settings, *[Paper]*
Alokendu Mazumder, **Manan Tayal**, R. Sabharwal, Bhartendu Kumar and Punit Rathore
Transactions on Artificial Intelligence (TAI).
Cyber Physical System Symposium (CyPhySS), 2024

2025 **[5] Semi-Supervised Policy Synthesis using Barrier Certificates**, *[Paper]*
Manan Tayal*, Aditya Singh*, Pushpak Jagtap and Shishir Kolathaya,
IEEE Conference on Decision and Control (CDC), Rio, Brazil, 2025.
IROS Workshop on *Formal Methods & Techniques in Robotic Systems*, 2024.

[4] A Physics-Informed Machine Learning Framework for Safe and Optimal Control of Autonomous Systems, *[Paper]*
Manan Tayal*, Aditya Singh*, Shishir Kolathaya and Somil Bansal.
International Conference on Machine Learning (ICML), Vancouver, Canada, 2025.

2024 **[3] Learning a Formally Verified Control Barrier Function in Stochastic Environment**, *[Paper]*
Manan Tayal, Hongchao Zhang, Pushpak Jagtap, Andrew Clark and Shishir Kolathaya.
IEEE Conference on Decision and Control (CDC), Milan, Italy, 2024.

[2] Control Barrier Functions in Dynamic UAVs for Kinematic Obstacle Avoidance: A Collision Cone Approach, *[Paper]*
Manan Tayal, Rajpal Singh, Jishnu Keshavan and Shishir Kolathaya.
American Control Conference (ACC), Toronto, Canada, 2024.

[1] Collision Cone Control Barrier Functions: Experimental Validation on UGVs for Kinematic Obstacle Avoidance, *[Paper]*
B.G. Goswami*, **Manan Tayal***, Karthik Rajgopal, Pushpak Jagtap and Shishir Kolathaya.
American Control Conference (ACC), Toronto, Canada, 2024.

Submitted Papers/Preprints

Under Review **[2] A Robust Physics-Informed Machine Learning Framework for Safe and Optimal Control of Autonomous Systems**, *[Paper]*
Manan Tayal*, Aditya Singh*, Ganga Nair, Shishir Kolathaya, Somil Bansal,
The International Journal of Robotics Research.(Submitted)

[1] MAD-PINN: A Decentralized Physics-Informed Machine Learning Framework for Safe and Optimal Multi-Agent Control, *[Paper]*
Manan Tayal*, Aditya Singh*, Shishir Kolathaya, Somil Bansal,
IEEE Robotics and Automation Letters (RAL).(Submitted)
AAAI Workshop on Multi-Agent Path Finding (MAPF), 2026.

Teaching Experience

Mar'24 – Present **Deep Learning for Robotics**, *Gujarat Technical University (GTU)*, Ahmedabad
- Delivered a series of invited lectures on modern deep learning techniques for robotic perception and control as part of the Prime Minister's Research Fellowship outreach.
- Designed course content, assignments, and hands-on demonstrations to introduce machine learning foundations for robotics applications.

May'24 – Jul'24 **E1 316o: Deep Learning for Robotics**, *Indian Institute of Science (IISc)*, Bangalore
- Head Teaching Assistant for the graduate-level course under Prof. Shishir Kolathaya and Prof. Pushpak Jagtap.
- Responsibilities included coordinating course logistics, supervising lab sessions, developing programming assignments, grading, and providing mentorship to student project groups.

Jan'24 – Apr'24 **CP315: Robot Learning and Control**, *Indian Institute of Science (IISc)*, Bangalore
- Head Teaching Assistant under Prof. Shishir Kolathaya.
- Led discussion sessions, assisted in syllabus structuring, designed evaluation rubrics, and guided students through reinforcement learning and control-oriented machine learning concepts.

Mar'23 – Oct'23 **Robotics and Reinforcement Learning**, *Manipal Institute of Technology*, Manipal
- Delivered invited lectures as part of the Prime Minister's Research Fellowship initiative.
- Designed and taught modules covering robotic systems, reinforcement learning fundamentals, and algorithmic implementation with practical demonstrations and Python-based assignments.

Invited Talks

2026 *"Towards Safe Foundation Models for Robotic Systems"*, at Microsoft Research India (Hosted by Dr. Akshay Nambi).

2025 *"Neural Safety Filters for Trustworthy Autonomy"*, at Ati Motors, Bangalore (Hosted by Dr. Naveen Arulselvan, CTO of Ati Motors).

2025 *"Towards Trustworthy Autonomy in Real World Environments"*, at Algorithmic Systems Theory (AST) Lab, ETH Zurich (Headed by Prof. Lars Lindemann).

2025 *"A Physics Informed Machine Learning approach for Safe and Optimal Control of Autonomous Systems"*, at Safe and Intelligent Autonomy (SIA) Lab, Stanford University (Headed by Prof. Somil Bansal).

2024 *"Learning Formally Verified Neural Control Barrier Functions in Stochastic Environments"*, at Cyber Physical Systems Symposium (CyPhySS), Bangalore.

2023 *"A Collision Cone Approach for Control Barrier Functions"*, at the Systems and Controls Workshop, IIT Bombay.

Research Mentoring

I have had the fortune of working with and mentoring some fantastic student collaborators.

2022-2024 GVS Mothish (M.Tech. @ IISc) → (Ph.D. @ CDS IISc)

2022-2024 Karthik Rajgopal (B.Tech. @ BITS Pilani) → (MEngg. @ UC Berkeley)

2023-2024 Ravi Kola (B.Tech. @ GTU) → (MS @ RWTH Aachen University)

2024-2026 Aditya Singh (B.Tech. @ IIT Patna)→ (PhD @ University of Pennsylvania)

2024-2025 Shreenabh Agarwal (BS @ IISc Bangalore)→ (MS @ Carnegie Mellon University)

Research Service

Peer Reviewing

Conferences International Conference on Machine Learning (ICML), IEEE International Conference on Robotics and Automation (ICRA), IEEE Conference on Decision and Control (CDC), Conference on Robot Learning (CoRL), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), American Control Conference (ACC), European Control Conference (ECC), Indian Control Conference (ICC)

Journals IEEE Transactions on Automatic Control (TAC), IEEE Robotics and Automation Letters (RA-L)

Session Co-chair

Conferences European Control Conference (ECC'24), International Conference on Control (ICC'25)